

# OpenTelemetry Demo

## System Architecture

High-Level View of Application Services, Telemetry  
Collection, and Observability

### Visual Documentation Case Study

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## Version History

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## 1. Overview

This visual documentation case study demonstrates how complex technical information can be analyzed, structured, and translated into a purposeful visualization. It presents the context and source material behind the visualization, the communication problem it addresses, the intended audience, and the design decisions used to determine its scope, structure, and level of detail.

The case study includes the resulting visualization and identifies the source material used to understand and represent the subject.



## 2. Case Study Fields

Each case study follows a consistent structure to document the context, communication need, intended audience, visualization approach, and supporting references behind the resulting visual solution.

Table 1 Case Study Fields

| Source Field                 | Description                                                                        |
|------------------------------|------------------------------------------------------------------------------------|
| <b>Source &amp; Context</b>  | What information or system was studied and where the visualization originated      |
| <b>Communication Problem</b> | What aspect of the information is difficult to understand or needs clarification   |
| <b>Audience</b>              | Who the visualization is intended to help                                          |
| <b>Design Decisions</b>      | Why a particular diagram type, scope, structure, and level of detail were selected |
| <b>Visualization</b>         | The resulting visual solution                                                      |
| <b>Source Material</b>       | The authoritative references used to understand and verify the subject             |

### 2.1. Source & Context

This visualization is based on the **OpenTelemetry Demonstration**, a distributed microservice application designed to demonstrate OpenTelemetry instrumentation and observability in a production-like environment.

The application contains services implemented in multiple programming languages that communicate using gRPC and HTTP.

### 2.2. Communication Problem

The complete demo contains numerous application and observability components, making the architecture potentially difficult for a newcomer to interpret. The visualization provides a simplified overview of the major functional groups, key service relationships, and the connection between application services and observability infrastructure.

## 2.3. Audience

Developers and technical team members who are encountering the OpenTelemetry Demo architecture for the first time.

## 2.4. Design Decisions

A **Component Architecture Diagram** is selected to show the major system components their functional groupings, and how do they interact.

The visualization emphasizes **system structure, component boundaries, and the relationships between major services**. Related microservices are grouped into functional areas to support a high-level understanding of the architecture.

To maintain focus on the primary architecture, the visualization intentionally omits:

- Detailed instrumentation configuration
- Feature-flag scenarios
- Testing components
- Lower-level service interactions
- Other implementation details not required to understand the primary architecture

## 2.5. Visualization

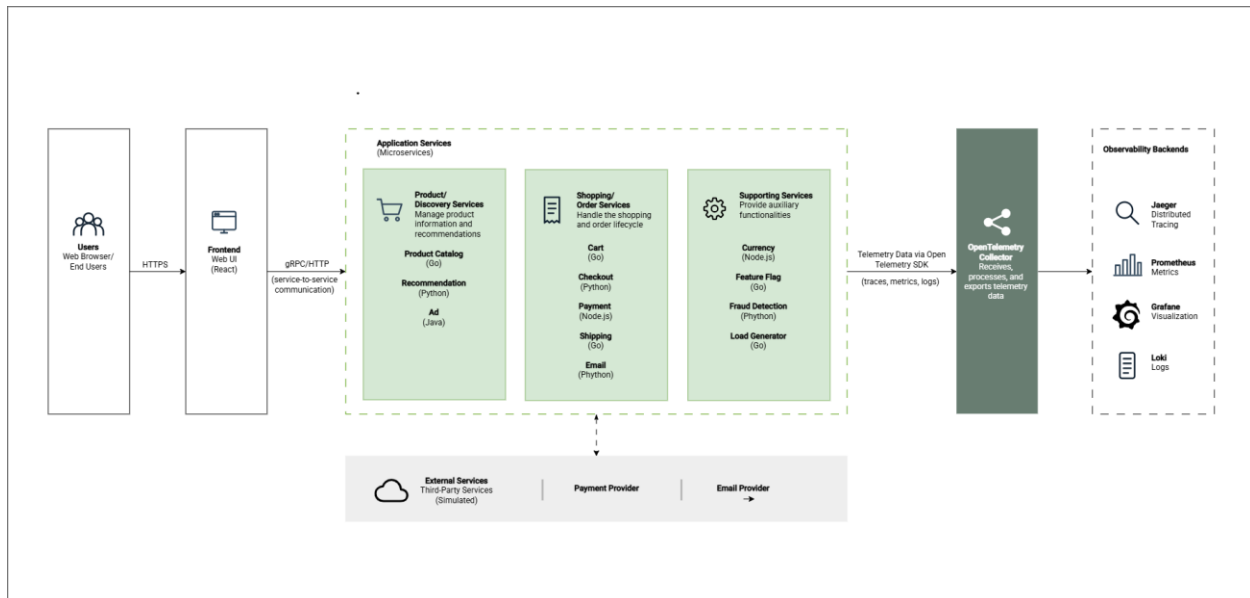


Figure 1 OpenTelemetry Demonstration

## 2.6. Source Material

The following are the links to the original public documentation/repository.

- [OpenTelemetry Demo - Architecture](#)
- [OpenTelemetry Demo - Services](#)
- [OpenTelemetry Demo - Architecture Requirements](#)